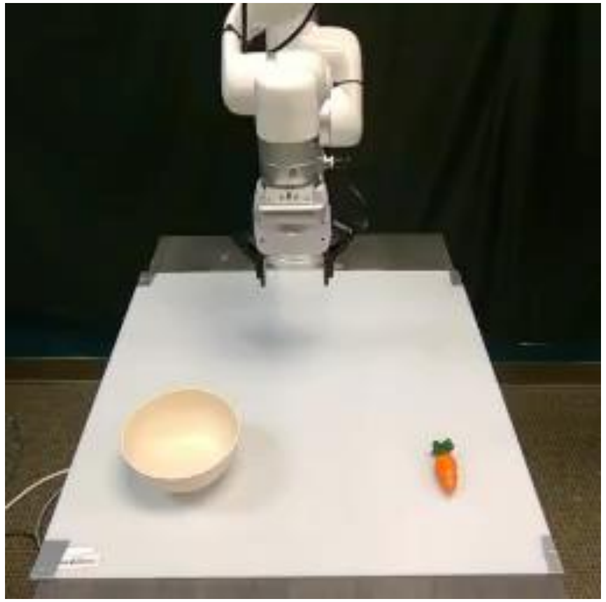
UniLACT: Depth-Aware RGB Latent Action Learning for Vision-Language-Action Models

Introduction

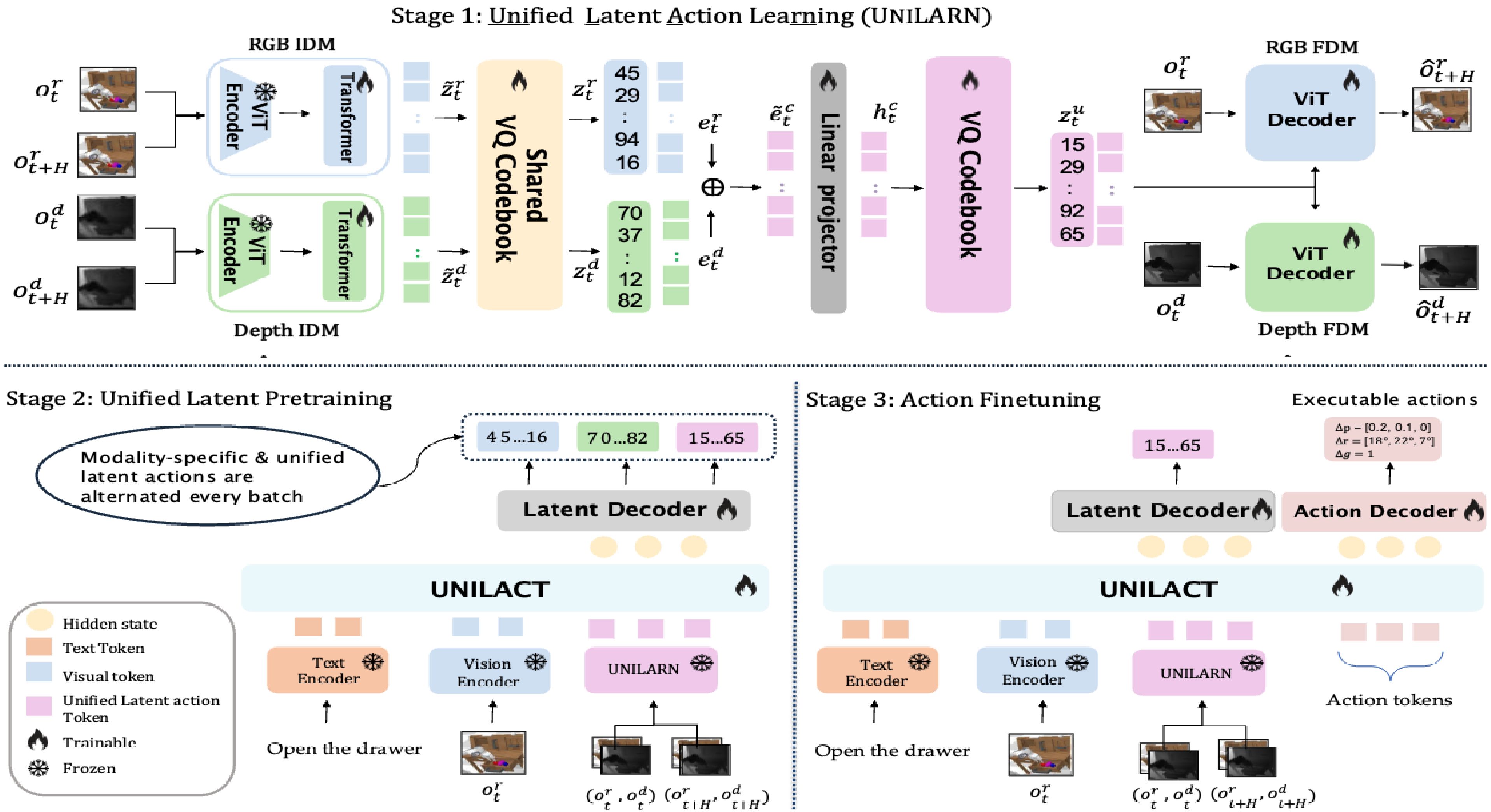
- Latent actions enable VLA pretraining from unlabeled videos.
- Existing latent action methods rely on RGB observations only with limited 3D geometry.
- 3D structure is critical for precise manipulation.



Fails to grasp the carrot due to inaccurate depth estimation

We introduce ***UniLACT****, a transformer-based VLA model that incorporates geometric structure through depth-aware latent pretraining.*

UniLACT Overview

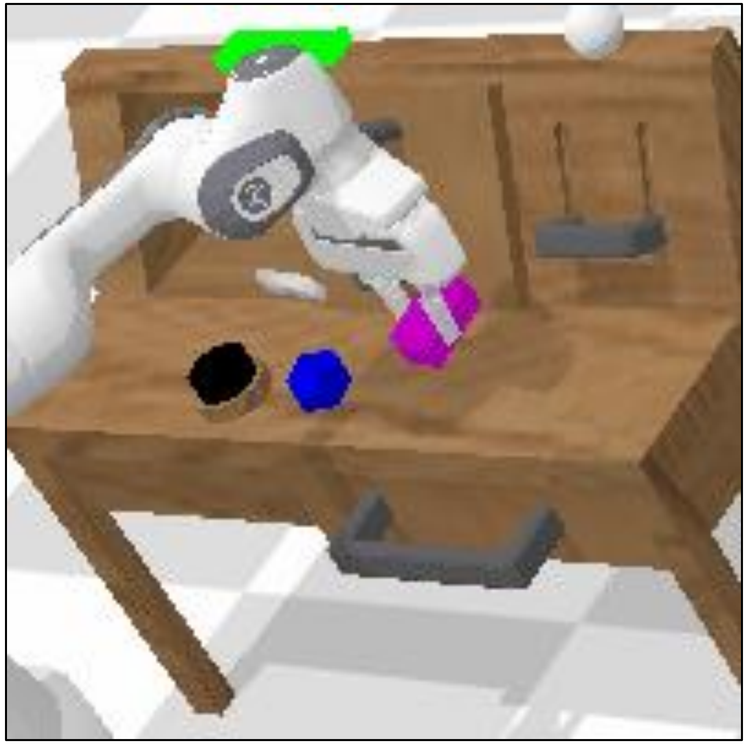


Results on CALVIN(ABC->D) Benchmark

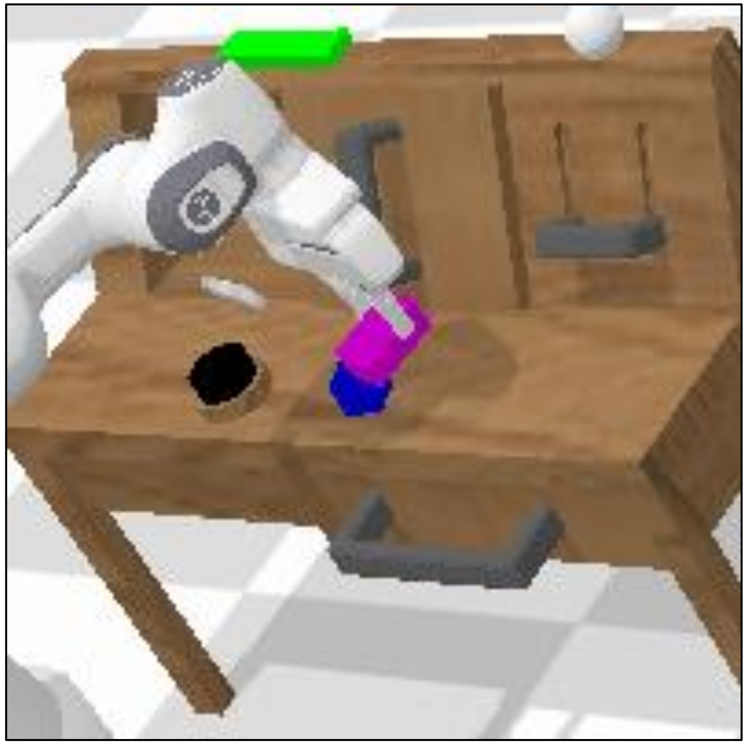
Model	Observation input	Tasks Completed in a row					Avg. Len ↑
		T1	T2	T3	T4	T5	
CALVIN(ABC) Pretraining							
Diffusion Policy[44]	Static RGB	0.402	0.123	0.026	0.008	0.00	0.56
HULC[45]	Static RGB	0.418	0.165	0.057	0.019	0.011	0.67
RT-1[46]	Static RGB	0.533	0.222	0.094	0.038	0.013	0.90
Robo-Flamingo[47]	(Static + Gripper) RGB	0.824	0.619	0.466	0.331	0.235	2.48
SuSIE[48]	Static RGB	0.870	0.690	0.490	0.380	0.260	2.69
GR-1[49]	(Static + Gripper) RGB + Proprio.	0.854	0.712	0.596	0.497	0.401	3.06
Moto[10]*	Static RGB	0.827	0.637	0.485	0.374	0.278	2.60
UniLACT (Ours)	Static RGB	**0.855**	**0.689**	**0.543**	**0.440**	**0.332**	**2.86**
OXE Pretraining							
Moto[10]*	Static RGB	0.824	0.60	0.439	0.316	0.228	2.40
UniLACT (Ours)	Static RGB	**0.897**	**0.729**	**0.584**	**0.493**	**0.401**	**3.10**

UniLACT outperforms the RGB-based latent action baseline by **29.2%.**

Simulation Rollouts (CALVIN)



Task: Store the grasped block in the sliding cabinet

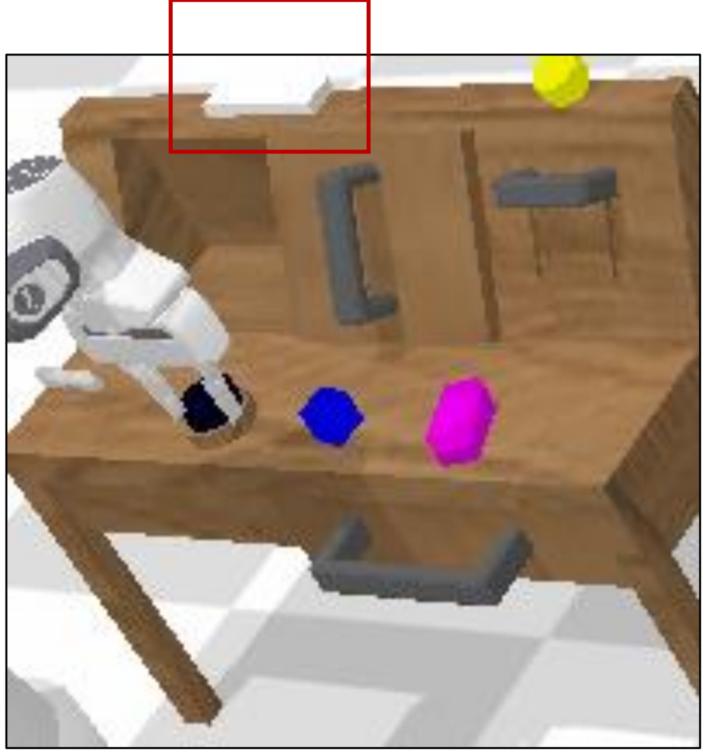


Moto ✗
(RGB latent action)

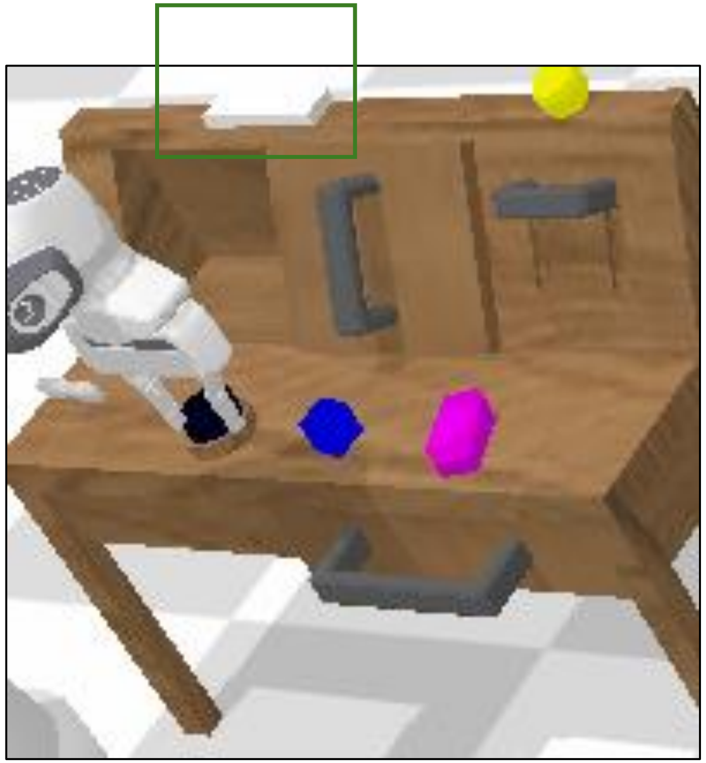
- Misjudges slider clearance
- Collides with the slider

UniLACT ✓
(RGB+Depth latent action)

- Maintains spatial clearance
- Collision-free placement



Task: Press the button to turn on the led light



Moto ❌
(RGB latent action)

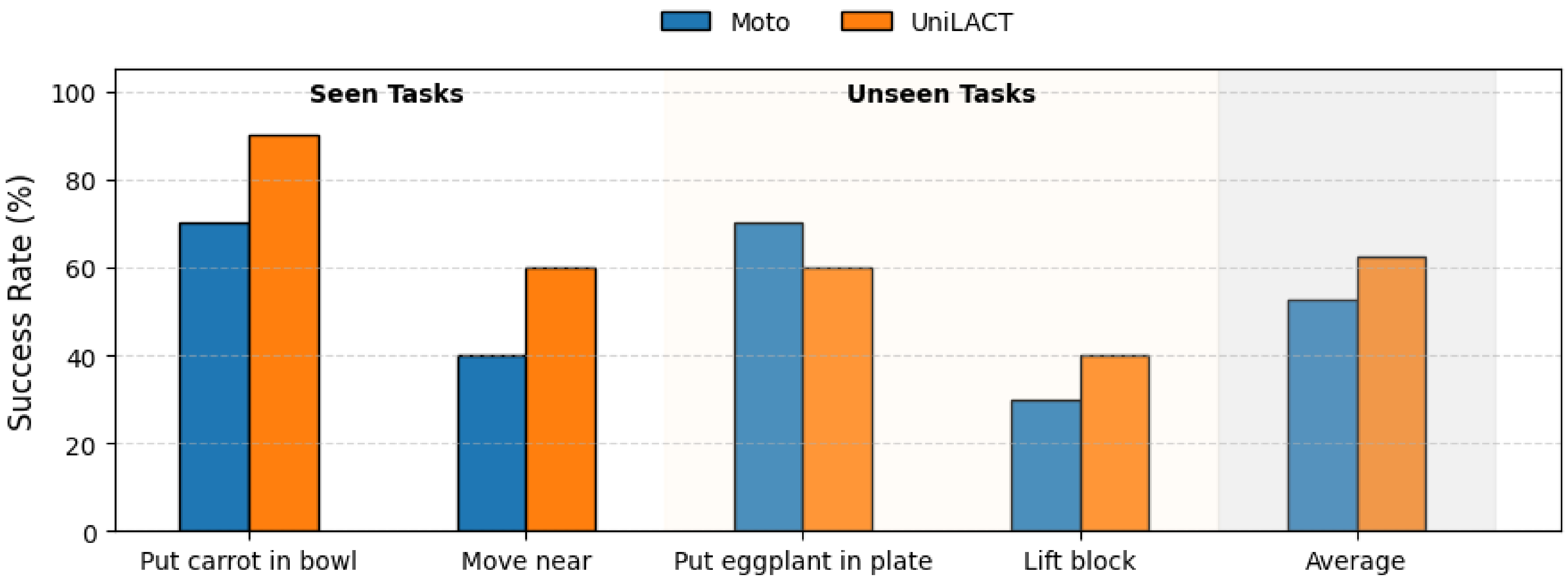
- Wrong estimation of button height
- Fails to press completely

UniLACT ✓
(RGB+Depth latent actions)

- Accurate button height positioning
- Successful button press

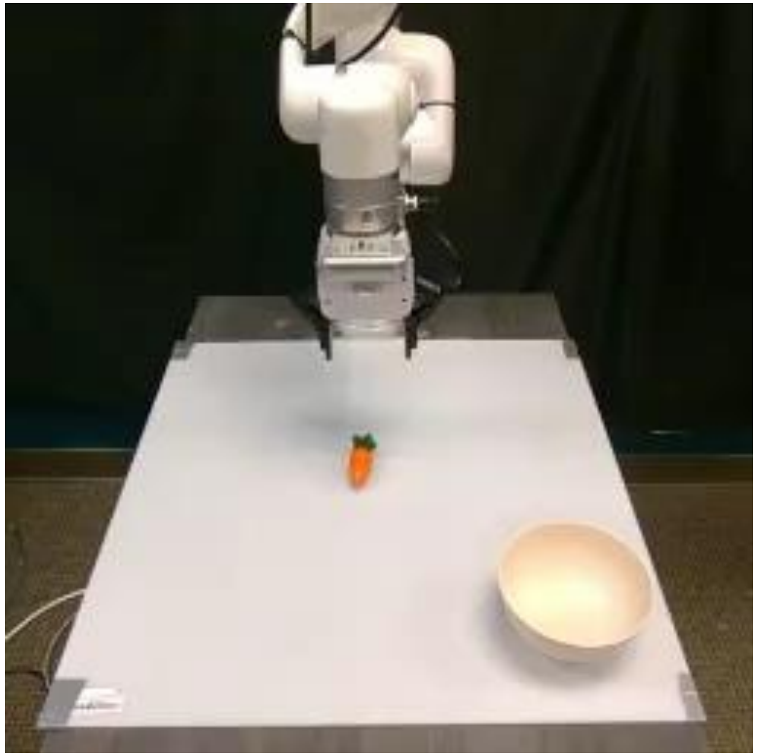
Integrating depth information in latent representations offers **strong geometric priors**

Real-world Results



Overall, **UniLACT** consistently outperforms the RGB latent baseline Moto on **3 out of 4 tasks** and achieves 10% performance improvements across all tasks.

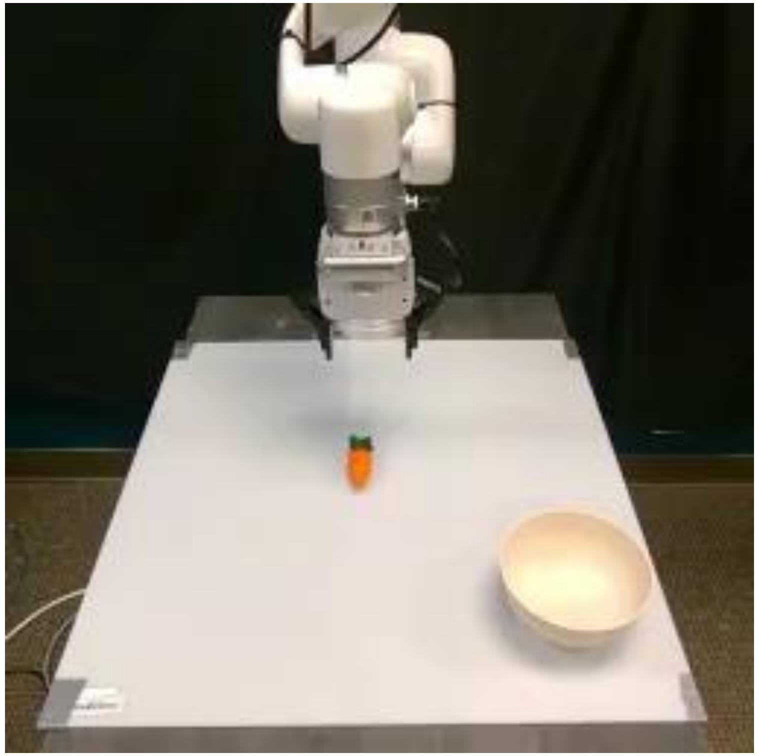
Real-world rollouts



Moto ❌
(RGB latent action)

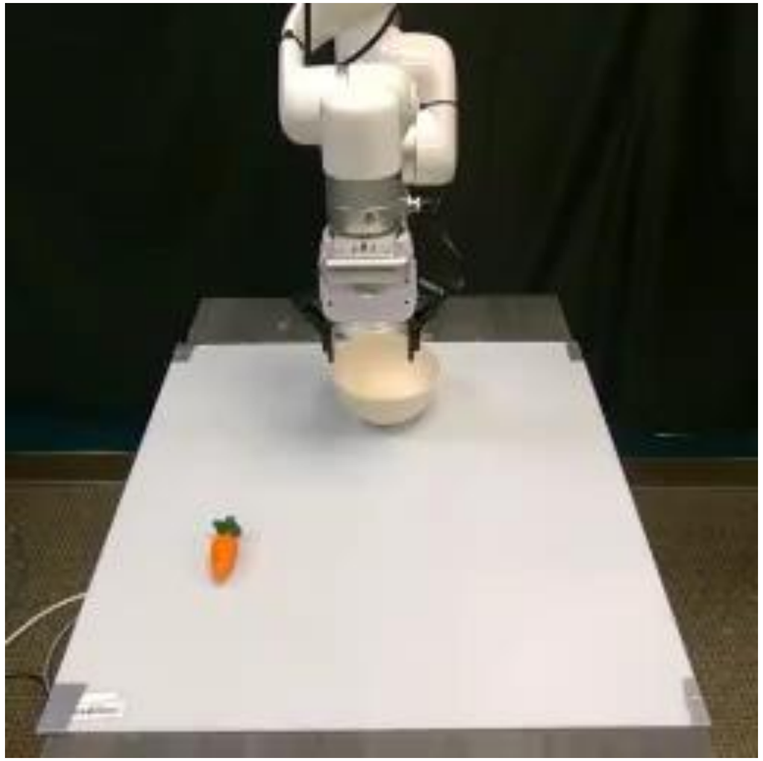
- Inaccurate depth estimation
- Collision with the bowl

Task: Pick up the carrot and place it in the bowl



UniLACT ✓
(RGB+Depth latent actions)

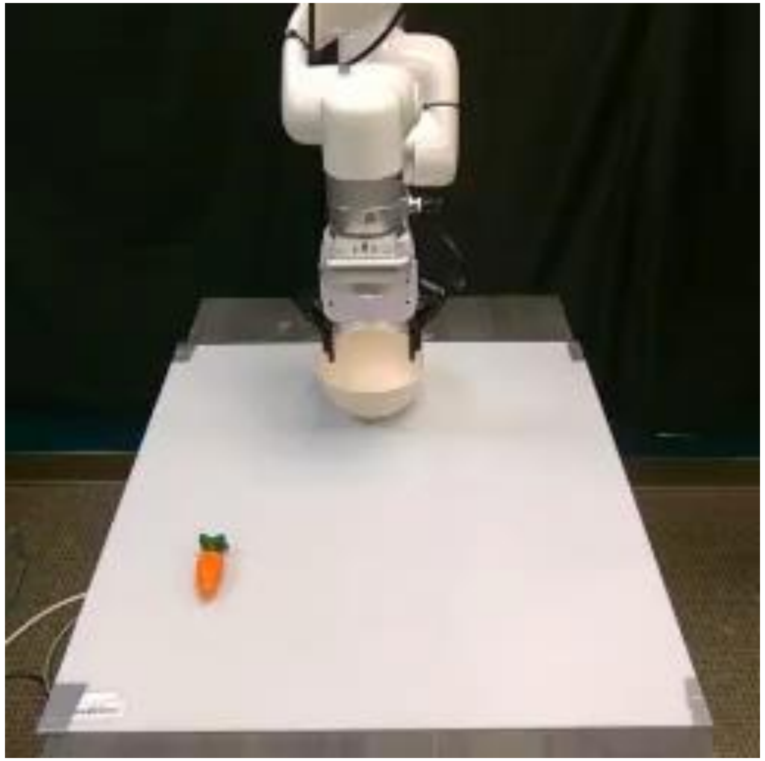
- Accurate depth estimation
- Avoids collision with the bowl



Moto ❌
(RGB latent action)

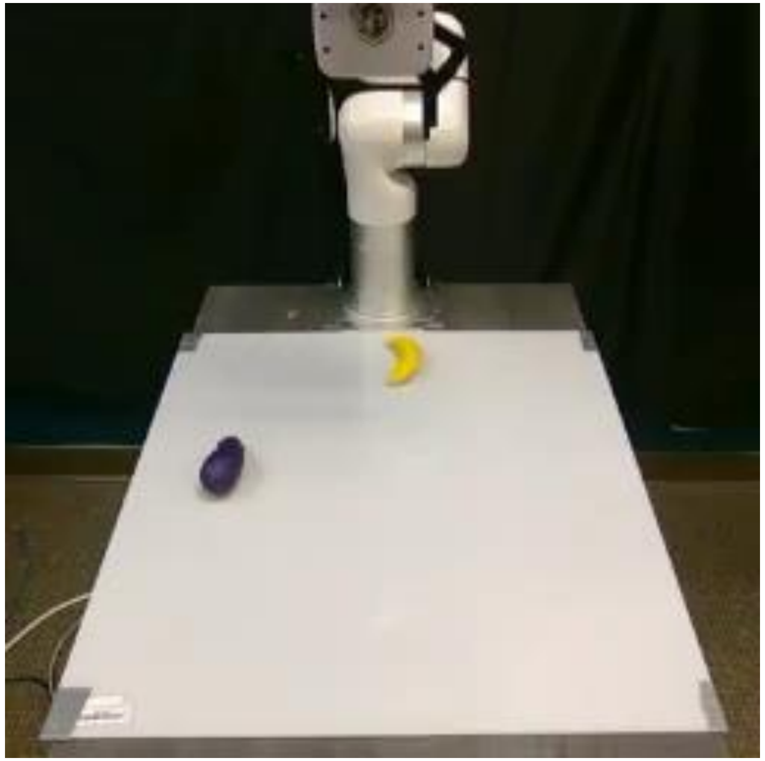
Fails to grasp

Task: Pick up the carrot and place it in the bowl

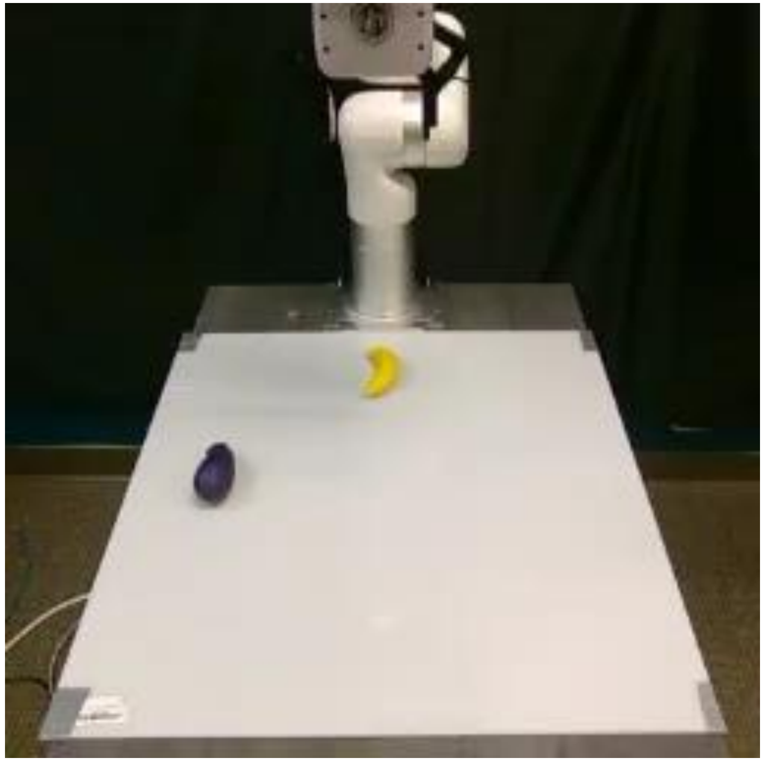


UniLACT ✓
(RGB+Depth latent actions)

Accurate 3D grasp



Task: Move the eggplant near the banana



Moto
(RGB latent action)

- Fails to grasp
- Collision while picking

UniLACT
(RGB+Depth latent actions)

- Accurate 3D grasp
- Collision-free pickup